Logarithmic scales

1 $y = \log_{10} x$

2

a 7 countries

b 3 countries

3

a 4 **b** 11 **c** 17 **d** 7

4

a 11 **b** 33 **c** 0.48 **d** 3 to 4

e 8

5

a \$79 432.82

b \$100 000 to \$1 000 000

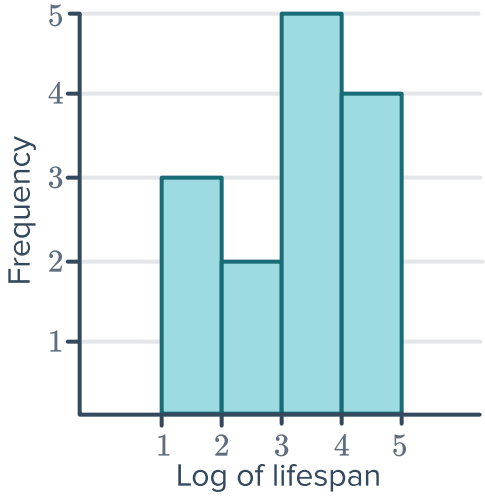
6

a

Animal/Insect	Lifespan (days)	Lifespan in a $\log_{10}$ scale
Ant	20	1.30
African elephant	25 500	4.41
Bowhead whale	71 400	4.85
Bumblebee	21	1.32
Chicken	2920	3.47
Dog	5110	3.71
Emperor penguin	8500	3.93
Fly	18	1.26
Giraffe	10 120	4.01
Hamster	780	2.89
Ladybug	365	2.56
Little penguin	2190	3.34
Monkey	7300	3.86
Shark	10 980	4.04

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b



Applications of logarithms

7

a $14\,000$

b $28\,000$

8

a $\$10\,000$

b $\$49\,000$

c less effective: every extra $\$1000$ spent on advertizing raises the sales revenue by less and less

9

a $R = \log\left(\frac{x}{a}\right)$

b

i $R = 3.8$ **ii** Minor

c

i $R = 4.2$ **ii** Light

10 3668 m

11

a 10 **b** 251 **c** 16 **d** 7.3

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12

a

i $127\ \text{dB}$ **ii** $42\ \text{dB}$

b 19.5

c $L = 80$

d $10 \log 2\ \text{dB}$

e

i $110\ \text{dB}$ **ii** $I = 10^{-\frac{57}{10}}\ \text{watts/cm}^2$

13

a 1000 **b** 10^7

c 1000 times louder **d** $1\,000\,000$

14

a $I = \dfrac{1}{10}$ MW **b**

i $(2, 3)$ **ii** $(6, 7)$

15

a 4 **b** 100

16

a 803 years

b 735 years

c R_1 : 1935 years, R_2 : 1771 years

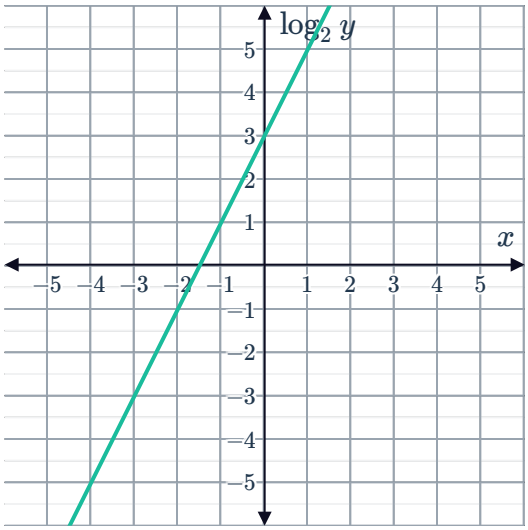
d No, at twice the half-life there will be one-quarter of the substance left. As the amount of substance approaches zero the time taken for a unit of substance to decay grows at an increasing rate.

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17

a $m = 2, k = 3$

b



c 2

d $x = \frac{1}{2}$

18 $\log \frac{15}{14}$

19

a 3.7 **b** Acidic **c** $h = 10^{-8.3}$

20

a **C** **b** 86 **c** 27 **d** 13 months

21

a $L = 7$ **b** $N = 62$ **c** $H = 41.68$ bits **d** $N = 16$

22

a 10 **b** $10 \log 5$ **c** $P = k$

Applications of exponential functions

23

a 19th day **b** $\$1\,597\,152$

24 $t = 99.02$ days

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25 $t = 1.39$

26

a 643

b $t = 8$

27

a $k = -\dfrac{\ln \dfrac{1}{2}}{5594}$

b $t = 11\,188.00$

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